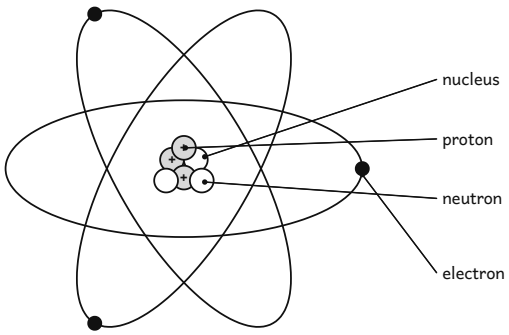
PHYSICAL AND CHEMICAL SCIENCE

Matter, Atoms and Conservation

Atoms and elements, conservation of mass in open and closed systems, and nuclear changes



In this pack

Learn it: Atoms: nothing is lost, everything changes ★☆☆
Fact check: Atoms and mass ★☆☆
Think it through: Where did the mass go? ★★☆
Investigate: Is mass conserved? ★★☆
Real-world case study: Energy from atoms: Rooppur ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · C3 · ATOMS AND CONSERVATION

Atoms: nothing is lost, everything changes

★☆☆
Grade 5

Name: Class: Date:

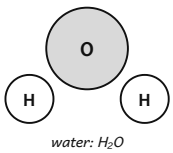
Learning goals

- Explain that matter is made of atoms and that atoms are conserved in physical and chemical changes.
- Explain why mass stays the same in a closed system.
- Describe simply how nuclear changes are different.

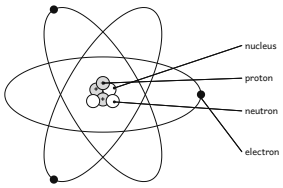
Everything around you (air, water, your desk, your body, the stars) is made of **matter**, and all matter is made of unbelievably tiny particles called **atoms**. A single human hair is about half a million atoms wide!

Atoms and elements

There are about 118 different kinds of atom. A material made of only one kind of atom is an **element**: oxygen, carbon, iron, gold and aluminium are elements. Atoms can join together to make **molecules**. A water molecule is two hydrogen atoms joined to one oxygen atom: **H_2O**.



water: H_2O



Each atom has a tiny centre called the **nucleus**, made of **protons** and **neutrons**, with even tinier **electrons** moving around it.

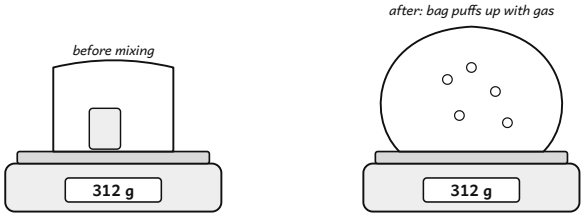
Atoms are conserved

In a **physical change** (melting, dissolving, evaporating) atoms just move around. In a **chemical change** (burning, rusting, cooking) atoms are **rearranged** to make new substances. But in both kinds of change, **no atoms are made and none are destroyed**. This is the **law of conservation of mass**: the total mass stays the same.

- 100 g of ice melts into 100 g of water.
- 250 g of water + 20 g of salt makes 270 g of salt solution.

Open and closed systems

Sometimes mass *seems* to disappear. When vinegar and baking soda react in an open cup, the mass goes down. That is because a **gas** (carbon dioxide) escapes into the air. If the reaction happens in a **sealed bag**, a **closed system**, nothing can escape and the mass stays exactly the same.



From the past

In the 1780s the French scientist **Antoine Lavoisier** weighed substances very carefully before and after reactions in sealed containers. He showed that mass is never lost, only changed. He is often called the father of modern chemistry.

Energy is conserved too

Energy cannot be made or destroyed either: it only moves from one place to another or changes form. In a closed system, the total amount of **energy** and **matter** stays the same.

Nuclear changes: atoms change

In a **nuclear** change, the nucleus itself changes. In a nuclear power station, big uranium atoms are **split** into smaller atoms of different elements, giving out a huge amount of heat. In the Sun, small hydrogen atoms **join** to make helium. So here atoms are **not** conserved: one element turns into another. But if you count all the **protons and neutrons**, the total before and after is the **same**.

Before	After
1 uranium atom: 236 protons + neutrons	barium (141) + krypton (92) + 3 free neutrons

141 + 92 + 3 = **236**. Nothing is lost!

Key words

matter	anything that has mass and takes up space
atom	the tiny building block of matter
element	a substance made of only one kind of atom
molecule	atoms joined together
conservation of mass	mass stays the same in a change
closed system	nothing can get in or out
nucleus	the centre of an atom
nuclear change	a change in the nucleus that turns one element into another

Atoms and mass

Name: Class: Date:

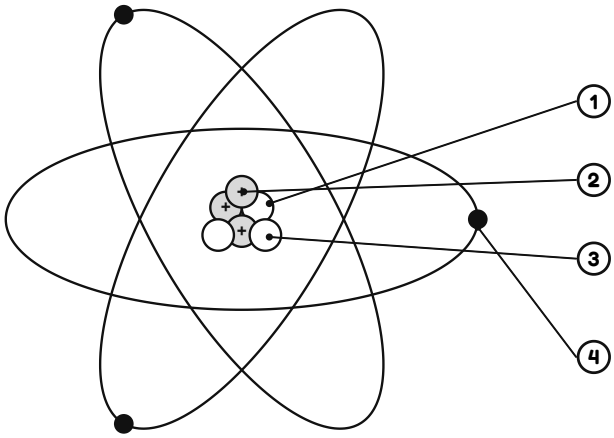
Learning goals

- Label an atom.
- Recall the law of conservation of mass.

1 Label the atom. [4]

LABELS

neutron nucleus electron proton



1 _____ 2 _____

3 _____ 4 _____

2 Fill in the gaps. [5]

WORD BANK

element closed bigger destroyed mass open atoms

All matter is made of tiny _____. A substance made of only one kind of atom is an _____. In physical and chemical changes, atoms are not made or _____, so the total _____ stays the same. A sealed bag is a _____ system.

3 Work out the missing masses. [3]

Change	Mass before	Mass after
ice melting	100 g of ice	
dissolving sugar	200 g water + 15 g sugar	
freezing juice	330 g of juice	

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. When sugar dissolves, its atoms are destroyed.			
2. Water is made of hydrogen and oxygen atoms.			
3. In a nuclear change, one element can turn into another.			
4. Burning wood makes the atoms disappear.			

Where did the mass go?

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Explain changes in mass using open and closed systems.
- Apply conservation to new examples.

1 **A candle has a mass of 50 g. After burning for an hour its mass is 42 g. Has mass been destroyed? Explain.** [2]

2 **Describe an experiment to show that mass is conserved when a candle burns.** [2]

3 **An iron nail gets heavier as it rusts. Where does the extra mass come from?** [2]

4 **Use the idea of conservation to check this nuclear change. Are the protons and neutrons conserved?** [2]

Before	After
a nucleus with 235 protons + neutrons, hit by 1 neutron	139 + 94 + 3 free neutrons

INVESTIGATE · C3 · ATOMS AND CONSERVATION

Is mass conserved?

★★☆
Grade 5

Name: Class: Date:

Learning goals

- Measure mass accurately.
- Compare open and closed systems.
- Explain unexpected results.

You will need

a digital balance, a zip-lock bag, a small cup, vinegar, baking soda, a teaspoon, an open plastic cup

Stay safe

Wear goggles. Vinegar can sting your eyes. Point the bag away from faces.

Class 5C carried out three tests and measured the mass before and after.

1 **Complete the table.** [3]

Test	Mass before (g)	Mass after (g)	Change (g)
dissolving salt in water (open cup)	270	270	
vinegar + baking soda in a sealed bag	312	312	
vinegar + baking soda in an open cup	312	309	

2 **Why did the mass go down in the open cup but not in the sealed bag?** [2]

3 **What evidence showed that a gas was made?** [1]

4 **Write a conclusion about conservation of mass.** [2]

5 **The balance shows whole grams only. How could this affect the results?** [1]

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REAL-WORLD CASE STUDY · C3 · ATOMS AND CONSERVATION

Energy from atoms: Rooppur

★★★ Grade 5

Name: Class: Date:

Learning goals

- Explain nuclear power simply.
- Weigh advantages and disadvantages.
- Consider different perspectives.

Science · Geography · Economics · Ethics · Environment

Bangladesh needs more and more electricity for homes, schools, hospitals and factories. Most of it comes from burning **natural gas**, which is running out and gives off carbon dioxide.

At **Rooppur**, in Pabna district on the bank of the Padma, Bangladesh is building its **first nuclear power plant**, with help from Russia, and it is starting up in the mid-2020s. Inside its reactors, uranium atoms are split. The heat boils water into steam, which spins turbines to make electricity. A small amount of uranium can produce a very large amount of energy, and the plant gives off almost no carbon dioxide while working.

But nuclear power also has challenges: it is very **expensive** to build; it must be run with great **safety** care; and the used fuel stays **radioactive** (dangerous) for thousands of years, so it must be stored very carefully.

Based on public information about the Rooppur Nuclear Power Plant.

1 **Put the steps of making electricity at Rooppur in order.** [3]

- ☐ Steam spins a turbine.
- ☐ Uranium atoms split in the reactor.
- ☐ The turbine turns a generator, making electricity.
- ☐ The heat boils water into steam.
- ☐ This releases a lot of heat.

2 **Complete the table.** [6]

Advantages of nuclear power	Disadvantages of nuclear power

3 **What might each person think about the plant?** [3]

Person	View
a factory owner in Dhaka	
a farmer living near Rooppur	
a climate scientist	

4 **Your opinion: should Bangladesh get more electricity from nuclear power, solar power or both? Give reasons.** [2]

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Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

A	T	O	M	V	S	V	T	H	J	Y
D	A	W	S	T	N	P	I	C	E	G
B	O	R	E	F	M	A	S	S	G	P
T	F	M	G	L	P	R	O	T	O	N
K	S	E	A	L	E	D	T	C	K	E
I	F	O	S	Y	T	M	Y	F	U	M
O	E	Y	N	U	C	L	E	U	S	D
E	L	E	C	T	R	O	N	N	L	L
R	H	K	M	J	S	I	A	W	T	A
C	M	U	M	O	L	E	C	U	L	E
Y	N	E	U	T	R	O	N	W	A	I

atom electron element mass molecule neutron nucleus proton sealed

2 **Salt (20 g) is dissolved in water (180 g). What is the mass of the solution?** [1]

- ◯ 160 g
- ◯ 180 g
- ◯ 200 g
- ◯ 220 g

3 **Which is a closed system?** [1]

- ◯ an open pan of boiling water
- ◯ a sealed jar
- ◯ a candle burning in a room
- ◯ a puddle drying

4 **Draw particle pictures to show that 10 atoms before a change are still 10 atoms after.** [2]

before

after

5 **Challenge. Explain why scientists say atoms are conserved in chemical changes but not in nuclear changes, and what is still conserved in a nuclear change.** [3]

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How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
explain that matter is made of atoms	○	○	○
explain why mass is conserved in physical and chemical changes	○	○	○
explain the difference between open and closed systems	○	○	○
measure mass before and after a change	○	○	○
explain how nuclear changes are different	○	○	○
discuss the advantages and disadvantages of nuclear power	○	○	○